32×32 Approximate CAM with Tunable Hamming Distance

Quick Guide

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What it does

The chip stores 32 words of 32 bits. Give it a search word and it tells you, for each row, whether that row is similar to the word (match) or not (mismatch). "Similar" means the number of differing bits does not exceed a threshold, and you program that threshold.

Pins

- `VDD`: 1.2 V. `VSS`: 0 V.
- `VEVAL_IN`: 0.5 V, the evaluation level (analog input).
- `CLK`: 100 MHz maximum; the chip works on the rising edge.
- `reset`: active high.
- `EN_WRITE`, `EN_REFERENCE`, `EN_SEARCH`: pulses that launch each operation.
- `serial_in_addr`: row address, 5 bits, most significant first.
- `serial_in_data`: 32-bit word, bit 31 first.
- `OUTPUT`: the result, one bit per row: 0 = match, 1 = mismatch.

Data are sent in normal form; the chip takes care of the polarity.

Power-up

1. Apply `VDD` = 1.2 V and `VEVAL_IN` = 0.5 V.
2. Start the clock.
3. Hold `reset` at 1 for at least 2 cycles, with every EN at 0, then release it.
4. Program the reference, write your words, and you are ready to search.

Basic rules

- Change inputs on the falling edge; the chip reads them on the rising edge.
- A 1- or 2-cycle EN pulse launches a single operation.
- The first rising edge with EN high is E0. Bit k of a serial input is read on edge E($k+1$).
- One operation at a time. The next one can start 35 cycles after a write or reference, or 70 cycles after a search. If sent earlier, it is ignored.

1. Program the threshold (once)

- Pulse `EN_REFERENCE`, then send the 32 bits of the reference word R on `serial_in_data`.
- Rule: **match if the differing bits are $m_{\mathbf{ref}} - 4$ or fewer**, where m_{ref} is the number of bits in which R differs from `0xFFFF0000`.
- Recommended: R = `0xFFF8FFFF` ($m_{\mathrm{ref}} = 19$): match with 15 differing bits or fewer, mismatch with 16 or more.
- For another threshold, start from `0xFFFF0000` and invert any m_{ref} bits.
- Takes 340 ns and is kept until you reprogram it or reset the chip.

2. Write a word

- Pulse `EN_WRITE`.
- Send the 32 bits on `serial_in_data` and, during the first 5 cycles, the address on `serial_in_addr` at the same time.
- Takes 340 ns.

3. Search

- Pulse `EN_SEARCH` and send the 32 bits of the search word on `serial_in_data`.
- The chip compares for 5 cycles and then shifts the result out on `OUTPUT` for 32 cycles: row 31 first and row 0 last.
- Row i comes out between edges $\mathrm{E}(68-i)$ and $\mathrm{E}(69-i)$. Read it on the falling edge.
- 690 ns in total. Rows you did not write give a meaningless value.

Suggested first test

Search for $Q =$ `0x0000FFFF` with reference `0xFFF8FFFF`, after writing:

Row	Word	Differing bits	Expected `OUTPUT`
0	`0x0000FFFF`	0	0
5	`0x0000E000`	13	0
13	`0x00FF01FF`	15	0
20	`0x00FF00FF`	16	1
22	`0x00030000`	18	1
31	`0xFFFF0000`	32	1

Calibration on the real chip

On silicon the threshold may shift by one or two bits. Write rows with 12 to 20 differing bits with respect to Q, search, find the row where `OUTPUT` changes from 0 to 1, and adjust m_{ref} one step at a time. Each unit of m_{ref} moves the threshold by about one bit. Keep `VEVAL_IN` fixed at 0.5 V; if you change it, calibrate again.